

# Kartik Patel

Senior Backend Engineer · Java · Spring Boot · Distributed Systems

[meetkartik40@gmail.com](mailto:meetkartik40@gmail.com) | +91 96852 56680 | [linkedin.com/in/demoralizer](https://www.linkedin.com/in/demoralizer) | [github.com/demoralizerr](https://github.com/demoralizerr) | [leetcode.com/frankenstrike](https://leetcode.com/frankenstrike)

8 years designing high-throughput microservices at Mastercard — 300–400 TPS event pipelines, Kafka-based distributed architecture, and cross-team platform initiatives that directly drove commercial revenue.

## EXPERIENCE

### Mastercard

Jan 2025 – Present | Pune, India

#### Senior Software Engineer

- Reduced file ingestion time by **67%** (1 hr → 20 min) for a 1 GB payload by redesigning the GET API to stream chunks directly into the database, eliminating in-memory buffering and improving system reliability under load.
- Spearheaded architecture and implementation of an Australian Business Number matching service (**60%+** of code-base) — designed the service boundary, API contract, and data flow, directly driving commercial payments revenue growth.
- Led technical design reviews for two cross-team integrations, defining integration patterns and data schemas that reduced downstream service coupling and improved long-term maintainability.

### Mastercard

Nov 2021 – Jan 2025 | Pune, India

#### Software Engineer II

- Designed a high-throughput transaction fragment merging pipeline processing **300+ TPS** inbound and publishing **400+ TPS** merged events — architected the consumer topology, concurrency model, and event schema for exactly-once processing guarantees.
- Increased Kafka topic onboarding speed by **90%** by designing a config-driven generic consumer framework, abstracting deserialization, error handling, and retry logic (days → hours per topic).
- Eliminated duplicate scheduled-job execution across **8** distributed instances by architecting a distributed locking solution (ShedLock) — defined transaction boundaries and lock TTL strategy; zero incidents post-deployment.
- Improved observability across the transaction pipeline by introducing structured logging and metrics instrumentation, enabling faster root cause analysis for production incidents.

### Tata Consultancy Services

June 2019 – Nov 2021 | Pune, India

#### System Engineer

- Designed a backend ETL microservice consuming **400+ TPS** real-time Kafka events — defined the data model, transformation logic, and deployment architecture on AWS via Docker for horizontal scalability.
- Improved test coverage from **10%** to **90%** across bill-pay backend services using JUnit — established testing patterns and mentored team members on test design, enabling confident CI/CD delivery.
- Established code review standards across a 6-engineer team — introduced structured feedback covering performance, correctness, and readability, cutting average PR cycle time by ~30% and reducing post-merge bugs.

## CODING ACHIEVEMENTS

**500+ problems solved (200+ Medium, 200+ Hard)** LeetCode · [leetcode.com/frankenstrike](https://leetcode.com/frankenstrike)

**Knight Badge** LeetCode · top 5% of competitive programmers globally

## SKILLS

**Core:** Java · Spring Boot · Apache Kafka · Microservices · REST API Design · Swagger/OpenAPI

**Data:** Oracle · SQL · Data Modelling · Query Optimisation · Hibernate (JPA)

**Reliability:** Distributed Locking · Concurrency · Transaction Management · JUnit (Unit + Integration)

**Observability:** Structured Logging · Metrics Instrumentation · Production Debugging · Root Cause Analysis

**Infrastructure:** PCF · AWS · Docker · Maven · Gradle · Spring Security

## EDUCATION

**BTech, Computer Engineering**

RGPV University | Aug 2015 – June 2019